

Project Based Learning (PBL) — Panel Presentation Content

Components 1–7 (Abstract through Results & Discussion)

ZONE — An Infrastructure-Free Bluetooth Store-Carry-Forward Network for Disaster Zones, with Silence-Based Damage Inference

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Native Android (Kotlin + Jetpack Compose) · minSdk 26 · 82 JVM unit tests · validated on three physical phones in airplane mode

This document contains the full content for PPT components 1–7. Each section is mapped to its slide budget from the PBL circular. All design diagrams are in Component 5; algorithm / pseudocode is in Component 6; result tables, comparison and discussion are in Component 7.

Component 1 — Abstract (Max. 1 slide)

Project summary, objectives, methodology and key outcomes.

- **Problem.** In the first 72 hours after an earthquake or flood, cellular and internet infrastructure is down or saturated. Trapped survivors cannot call for help, and incident command cannot tell a collapsed block from a lightly-flooded one.
- **Objective.** A single native-Android application that (i) forms an infrastructure-free mesh between ordinary smartphones over Bluetooth Low Energy (BLE), and (ii) infers where damage is worst from the pattern of devices that fall silent.
- **Methodology.** Connectionless dual-PHY BLE advertising (1M + Coded); a fixed 31-byte packet; epidemic store-carry-forward relaying with content-addressed de-duplication (a CRDT by construction); a battery-adaptive duty cycle; and a novel "Dead Man's Packet" evaluator in which every packet declares its own next transmission time, so that silence becomes a measurable signal — expected (dying battery, announced) versus unexpected (promised to speak, then vanished = destroyed).
- **Key outcomes.** One APK with three role-specific UIs (Citizen / Responder / Map-EOC); 82 JVM unit tests, all passing, run in CI on every push; validated on three physical phones in airplane mode — 2-hop message delivery with the source's Bluetooth switched off, correct discrimination of expected versus unexpected silence, and detection of simultaneous cell-wide silence ("CELL_LOSS") reproduced against a 500-node simulation of the 2015 Gorkha earthquake casualty curve (22 → 95 → 469 → 626 deaths over 72 h).

Component 2 — Introduction (Max. 1 slide)

Background of the topic, context, motivation and significance of the work.

- **Disasters sever telecom.** In the 2015 Gorkha (Nepal) earthquake, communications, power and roads failed together; coordination between district and central emergency operations centres was ineffective because of "a weak database and an absence of modern technology," while roughly 1.4 million people needed food, water and shelter within days.
- **The golden 72 hours.** Survival probability for trapped victims falls sharply after about 72 hours; rescue effort is squandered when responders cannot localise victims or rank sites by need.

- **Existing tools do not fit.** Satellite phones are scarce and need a sky view. Mesh-messenger apps (Bridgefy, FireChat) are built for chat between two conscious users and have documented security failures. None of them treats a device going offline as information — yet that is the single most important event in a collapse.
- **Motivation and significance.** Every modern phone already carries the radio (BLE 5), the sensors (barometer, accelerometer, GNSS) and the battery to be a disaster node. The missing piece is software that needs no infrastructure and no user action, and that turns the absence of signal into a live damage map. Zone converts "search this whole block" into "search this 100 m cell — 12 phones went silent together at 04:12."

Component 3 — Literature Review (Max. 2 slides)

Paper name and publication details, inferences, relevance and research gaps (seven research papers).

3.1 DTN routing and mesh foundations

#	Paper / Publication	Key inference	Relevance to Zone	Research gap
1	A. Vahdat, D. Becker, "Epidemic Routing for Partially Connected Ad Hoc Networks," Tech. Report CS-2000-06, Duke University, 2000.	Flood every message to every contact; nodes exchange summary vectors and pull what they lack. Guarantees eventual delivery under intermittent connectivity.	Zone's relay is epidemic flooding — the right choice when contact patterns are chaotic and unpredictable.	Unbounded replication drains battery and buffers; no dedup identity is defined for a real byte-level format.
2	A. Lindgren, A. Doria, O. Schelén, "Probabilistic Routing in Intermittently Connected Networks" (PRoPHET), ACM MobiHoc 2003 (poster); IRTF RFC 6693, 2012.	Use history of encounters plus transitivity to forward selectively — more deliveries than epidemic at lower overhead.	Confirms the trade-off Zone must manage: replication versus cost.	Needs stable, repeated encounters to learn from — absent in the first hours of a disaster; assumes trustworthy peers.
3	Z. Lu, G. Cao, T. La Porta, "TeamPhone: Networking Smartphones for Disaster Recovery," IEEE Trans. Mobile Computing, 16(12):3554–3567, 2017.	Bridges cellular + ad-hoc + opportunistic networks; a "self-rescue" module groups and positions trapped survivors' phones; about 40% lower transmission energy.	Validates smartphone-only disaster networking and the idea of a survivor-side sub-system.	Positioning needs GPS or manual grouping; no notion of inferring damage; requires user interaction.
4	S. M. Darroudi, C. Gomez, "Bluetooth Low Energy Mesh Networks: A Survey," Sensors (MDPI), 17(7):1467, 2017.	BLE's native topology is star / point-to-point with short range; multi-hop mesh needs an added protocol layer.	Justifies Zone's connectionless advertising flood (no GATT, no pairing) as a robust mesh substrate.	Surveyed schemes assume GATT connections or the heavy standard BLE-Mesh stack — fragile under churn.

3.2 Radio range, sensing and security

#	Paper / Publication	Key inference	Relevance to Zone	Research gap
5	A. R. Sheikh et al., "Adaptive Physical Layer Selection for Bluetooth 5: Measurements and Simulations," <i>Wireless Communications and Mobile Computing</i> (Wiley/Hindawi), vol. 2021, Art. 8842919.	Measured BLE 5: 1M PHY $\approx$ 655 m, Coded PHY $\approx$ 1300 m line-of-sight at 0 dBm; realistic urban $\approx$ 300 m. Adapting PHY to conditions improves reliability.	Directly motivates Zone's battery→PHY ladder (Coded when affordable for reach, 1M in survival mode).	No link to duty-cycle or energy budgeting; no disaster application.
6	W. Falcon, H. Schulzrinne, "Predicting Floor-Level for 911 Calls with Neural Networks and Smartphone Sensor Data," arXiv:1710.11122, 2017.	Barometric pressure change from building entry gives floor level — 100% correct across 63 tests in 5 New York towers, with no beacons or building maps.	Grounds Zone's relative-altitude signal (basement versus rooftop; rising trend = drowning).	Needs a clean "entered building" event from GPS; inference is centralised at a call centre, not on the mesh.
7	M. R. Albrecht, J. Blasco, R. B. Jensen, L. Mareková, "Mesh Messaging in Large-Scale Protests: Breaking Bridgefy," <i>Topics in Cryptology – CT-RSA</i> 2021, LNCS 12704, Springer.	Bridgefy: plaintext sender/receiver IDs (tracking), no authenticity, trivial impersonation from a sniffed BLE handshake, whole-network denial of service from one malformed message.	Sets Zone's security bar: per-install hashed identity, auth-shape check on every relay, content-addressed dedup that drops malformed / duplicate frames, Sybil-resistant scoring, a pre-shared responder key.	Shows current disaster/mesh apps are unsafe; no work on making the inference layer robust to spoofed nodes.

3.3 Consolidated research gap

Delay-tolerant routing is solved in theory but assumes cooperative, trackable peers; disaster smartphone systems still require user action and GPS; BLE mesh work ignores battery-honest duty cycling; and no prior system treats device silence as a first-class sensing signal for damage assessment. Zone targets exactly this intersection.

Component 4 — Problem Identification (Max. 1 slide)

Clearly defined problem statement from the existing limitations / gaps, and the need for the proposed work.

Limitations of the current state of the art

- Infrastructure dependence and user burden.** Existing tools need a tower, a satellite, dedicated hardware, or two conscious users actively messaging. A person who is unconscious or buried, or a command post with a dead uplink, gets nothing.
- Silence is discarded.** Every DTN / mesh design treats a node dropping off as noise. In a collapse it is the signal — but only if each node has promised when it would speak next, which no packet format currently carries.
- No trustworthy situational picture.** Reports are fragmented, contradictory and unverified (Bridgefy-class spoofing); responders cannot rank sites or localise victims vertically.

Problem statement

Design and build an infrastructure-free, action-free communication and situational-awareness system for the first 72 hours of a disaster, that (a) relays a compact distress signal between ordinary smartphones over BLE with no pairing and no servers; (b) makes every packet declare its next transmission so that silence can be classified as expected or catastrophic; (c) detects area-wide destruction from correlated silence; and (d) stays useful and trustworthy on a dying, possibly hostile, network.

Need for the proposed work

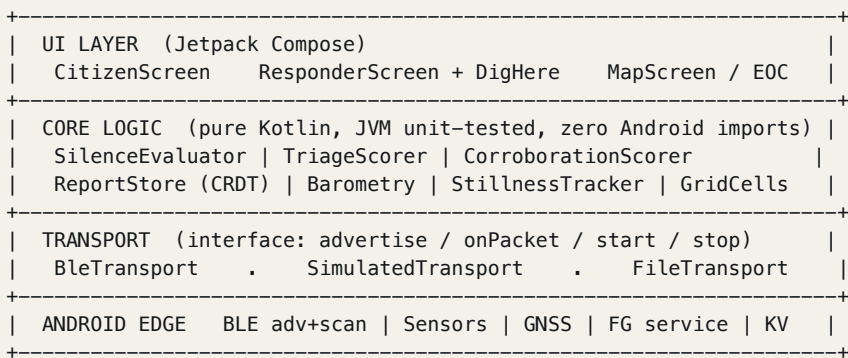
Convert "search this whole block" into "search this 100 m cell — 12 phones went silent together at 04:12," with a confidence-scored map and a vertical (floor) triage axis that GPS cannot provide.

Component 5 — Solution Strategy (Max. 3 slides)

Proposed approach, methodology, system design / architecture and planned solution. All design diagrams are included here.

5.1 System architecture and the three roles

Diagram 1 — Layered architecture:



- Transport is an interface (BLE / 500-node simulator / JSON file) so every algorithm runs identically on any of them — if BLE fails, the demo still runs on the simulator.
- The core/ package has no Android imports and is fully unit-testable on the JVM.
- A foreground service (connectedDevice type) keeps the radio alive when the screen is off.

Diagram 2 — One binary, three physical contexts:

CITIZEN	RESPONDER	MAP / EOC
dark, one number	sunlight, gloved, one hand	projector-facing
"You are being	triage list sorted by need	100 m severity grid
heard / 3"	+ "Dig Here" proximity bar	+ silence layer + replay
panicking user	metal-detector, not a map	CELL_LOSS is visible

5.2 Data path: store-carry-forward and the Dead Man's Packet

Diagram 3 — The relay (a message crosses a gap on foot):

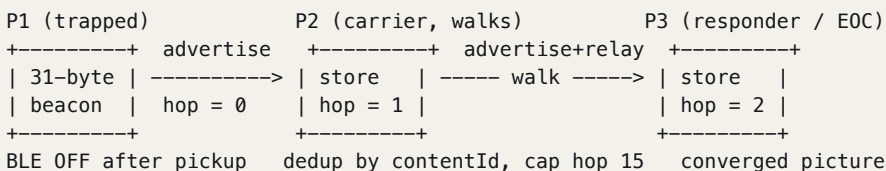


Diagram 4 — Dead Man's Packet state machine (maintained per known device):

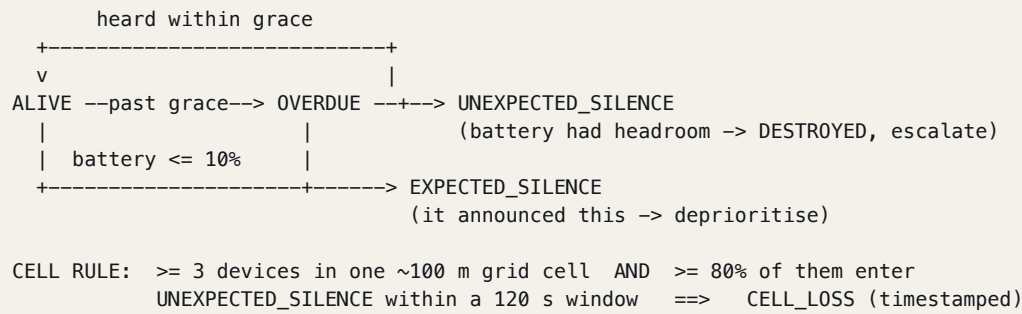


Diagram 5 — Battery-driven duty-cycle ladder:

Battery	Interval	PHY	next_expected_tx broadcast
> 60%	1 s	Coded (maximum reach)	1
30–60%	10 s	alternate Coded / 1M	10
10–30%	60 s	1M only	60
< 10%	300 s	1M, sparse	300 (still honest)

5.3 Packet format, de-duplication and trust model

Diagram 6 — The 31-byte packet:

Off	Len	Field	Notes
0	1	version + type	type 0 = STATUS, 1 = RESOLVE
1	6	device_id	first 6 B of SHA-256(per-install random key)
7	4	position	lat/lon delta $\times 10^5$ from a per-incident origin; sentinel = no fix
11	1	status	UNKNOWN / SAFE / TRAPPED / RISING_WATER / INJURED / RESOURCE / RESPONDER
12	1	severity · casualties	4 bits each
13	2	timestamp	minutes since event epoch
15	1	battery · hops	4 bits each
16	2	next_expected_tx (s)	the field the USP runs on
18	1	alt_delta	signed metres versus baseline; sentinel = no barometer
19	4	auth	first 4 B of SHA-256(key payload) — anti-spoof, not full crypto
23	1	alt_trend	metres over last 3 transmissions (+ = climbing)
24	7	reserved	a RESOLVE packet carries the target's content-id prefix here

Diagram 7 — CRDT de-duplication:

```
contentId = SHA-256( payload with hop nibble masked, auth bytes excluded )
-> the same message heard at any hop count is ONE record
-> merging two stores is set-union over contentId = conflict-free, no library
```

Trust model:

- Hashed per-install identity; the auth shape is verified on every relay.
- Malformed and duplicate frames are dropped at ingestion.

- CorroborationScorer weights each cell by distinct devices, independent hop paths, physical plausibility and a key-verified RESPONDER report.
- A fabricated-ID flood cannot inflate a cell (Sybil-resistant); an optional path-diversity guard protects CELL_LOSS itself.

Component 6 — Implementation (Max. 3 slides)

Technologies / tools used, development methodology, implementation details and major modules (algorithm / pseudocode).

6.1 Technologies, tools and methodology

Area	Choice
Language / UI	Kotlin 1.9, Jetpack Compose (Material 3), single-Activity
SDK	minSdk 26, targetSdk 34, compileSdk 34; AGP 8.7, Gradle 8.9
Radio	BluetoothLeAdvertiser + BluetoothLeScanner; manufacturer-specific data (company ID 0xFFFF); AdvertisingSetParameters for 1M and LE Coded (S=8); ScanSettings legacy=false + PHY_LE_ALL_SUPPORTED
Sensors	TYPE_PRESSURE (relative altitude, EMA-smoothed); TYPE_LINEAR_ACCELERATION (immobility); LocationManager GPS / PASSIVE — no Play Services, no network
Persistence	SharedPreferences (identity, config) + atomic JSON snapshot of the store and detected collapses in filesDir
Lifecycle	Foreground service (connectedDevice type); coalesced state pump at ≤ 4 Hz
Testing / CI	JUnit4, 82 JVM unit tests; GitHub Actions runs the tests plus assembleDebug on every push
Hard constraints	No network calls, no crypto library, no downloaded map tiles — airplane-mode native
Methodology	Phased, checkpoint-gated build H0→H8 (capability probe → codec + tests → advertiser / scanner → store / relay → Dead Man's Packet → barometer → three UIs → 500-node simulator → three-phone field validation), then tiered hardening

6.2 Core algorithm (pseudocode)

```
# --- build this device's heartbeat (every duty-cycle tick) ---
function BUILD_HEARTBEAT(now):
    batt      = effectiveBatteryPercent()
    interval  = LADDER_INTERVAL(batt)           # 1 / 10 / 60 / 300 s
    phy       = LADDER_PHY(batt)                # Coded / alternate / 1M
    fix       = freshGpsFix(maxAge = 10 min)    # else NO_FIX
    status    = userAssertion()
                or (RESPONDER if hasResponderKey and mode == RESPONDER)
                or (RISING_WATER if altimeter.rising)
                or (TRAPPED_DEBRIS if stillness.isStill(now, 5 min))
                or UNKNOWN
    pkt       = Packet(v=1, type=STATUS, deviceId, deltaPos(origin, fix),
                      status, severity, casualties = SelfReport.headcount,
                      stampMinutes(now), battNibble, hop = 0,
                      nextExpectedTx = interval,
                      altDelta = altimeter.deltaByte(), altTrend = altimeter.trend())
    bytes     = ENCODE_31B(pkt)
    signingKey = responderKey if status == RESPONDER else deviceKey
    bytes[19..22] = SHA256(signingKey || bytes[0..18])[0..3]
    ADVERTISE(bytes, phy)

# --- on every received advertisement ---
function ON_PACKET(bytes, rssi, tRx):
    if len(bytes) != 31 or not AUTH_SHAPE_OK(bytes): return
    cid = SHA256( maskHopAndDropAuth(bytes) )
    if STORE.contains(cid):
        STORE.fold(cid, rssi, tRx); return          # dedup: CRDT no-op
    STORE.put(cid, bytes, tRx)                      # new record
    pkt = DECODE(bytes)
    if pkt.deviceId != myId:
        SILENCE.onPacket(pkt.deviceId, pkt, tRx)
    if IS_RESOLVE(bytes) and VERIFY(bytes, responderKey):
        RESOLVE_LOG.add(prefix(bytes))              # mark that report handled

# --- relay pump (round-robin; never starves this device's own signal) ---
every PUMP_PERIOD (2 s):
    result = SILENCE.tick(now)
    for transition in result: LOG(transition, wallClock(now))
    if pumpTick mod 3 == 0:
        p = STORE.nextCarried(skip = devices believed EXPECTED / UNEXPECTED silent)
        if p and hop(p) < 15: ADVERTISE( INCREMENT_HOP(p) ) # copy; original untouched
    else:
        ADVERTISE( BUILD_HEARTBEAT(now) )

# --- Dead Man's Packet: classify one device (each tick) ---
function RECLASSIFY(track, now):
    promised = track.nextExpectedTx * 1000
    grace    = max(8 s, 2 * promised)
    overdue  = now - (track.lastHeard + promised)
    next     = ALIVE           if overdue < grace
                = EXPECTED_SILENCE if track.battery <= 10
                = UNEXPECTED_SILENCE if overdue > 3 * promised
                = OVERDUE         otherwise
    if next != track.state:
        LOG(track.id, track.state, "->", next, wallClock(now))
        track.state = next
        track.unexpectedSince = now if next == UNEXPECTED_SILENCE else null

# --- area destruction from correlated silence ---
function DETECT_CELL_LOSS(now):
    for (cell, members) in tracks grouped by ~100 m grid cell:
        silent = members where unexpectedSince != null
        if |members| >= 3
            and |silent| >= 0.8 * |members|
            and (max(silentSince) - min(silentSince)) <= 120 s
            and cell not already flagged:
```

```
FLAG CELL_LOSS(cell, |members|, |silent|,  
               firstSilent, lastSilent, detectedAt = now)
```

6.3 Major modules

Module	Responsibility
PacketCodec	encode / decode 31 B, contentId, incrementHop, verifyAuthWithKey, buildResolve
ReportStore	content-addressed set, dedup, lowest-hop-wins, round-robin relayBatch, union mergeFrom
SilenceEvaluator	Dead Man's Packet — per-device state, timestamped transition log, detectCellLoss
TriageScorer	single rank: rising-water+climbing → just-lost → trapped-below-grade → about-to-go-silent → severity
CorroborationScorer	0–1 cell confidence from distinct devices, hop-path diversity, plausibility, key-verified responder
Barometry / RelativeAltimeter	pressure → smoothed relative altitude + trend; graceful no-barometer
StillnessTracker	linear-acceleration magnitude → sustained immobility → inferred TRAPPED
SimulatedTransport	500 nodes on Rasuwagadhi geography, scripted mass-silence event
TransportController / BleForegroundService	keep the radio alive; own the store and evaluators; drive the pump; live EOC auto-export

Component 7 — Results and Discussion (Max. 3 slides)

Results obtained, analysis of findings, comparison where applicable, and discussion of outcomes (tabular form, graphs, screenshots).

7.1 Functional validation — three real phones, airplane mode

Devices: Samsung Galaxy S23 (trapped), Realme 7 (carrier), Motorola Edge 50 Pro (responder). Single-phone re-check on the Realme 7 on 2026-09-04.

#	Test	Result	Evidence
T1	Direct link, 1 hop	PASS	Responder shows both phones, 1 hop, LIVE
T2	Store-carry-forward (the demo)	PASS	S23 with Bluetooth OFF after pickup; its TRAPPED signal still reached the responder at 2 hops, relayed by the carrier
T3	Unexpected silence	PASS	Healthy battery, powered off → row turns SILENT (red) with the wall-clock time it went dark
T4	Expected silence	PASS	8% battery (declares 300 s), powered off → EXPECTED (grey), no escalation — the two are told apart
T5	Barometric floor	PASS (±2 m)	one floor read +2 m (spec ~±1 m); sign and magnitude correct
T6	Drowning escalation	PASS	"Water rising" + climbing altitude → jumps to top of the triage list
T7	Dig Here proximity	PASS	bar and WARMER / COLDER trend track walking toward and away
T8	CELL_LOSS	PASS	500-node simulator — one cell flips to a hatched, timestamped "12 dark" hole
—	Coded PHY reception	PASS	rxByPhy = {CODED, ONE_M} confirmed on the Motorola
—	Capability probe (H0)	—	S23 / Realme 7 / Motorola: isLeCodedPhySupported, isLeExtendedAdvertisingSupported, isMultipleAdvertisementSupported all true; max adv data 192 B; barometer present only on the S23

7.2 Quantitative results and comparison

Table A — engineering results (this build):

Metric	Value
Packet size	31 B core (BLE legacy-safe) — "a life in 31 bytes"
Unit tests / CI	82 tests, 100% pass, on every push
Simulator scale	500 nodes; Rasuwagadhi toll curve 22 → 95 → 469 → 626 over 72 h, replayed in ~90 s, no dropped frames
Duty-cycle range	1 s → 300 s; PHY Coded → 1M as battery falls
CELL_LOSS latency	detected within the 120 s correlation window; transition timestamped to the second
Heartbeat on Realme 7 @ 27%	interval 60 s, PHY 1M, next_expected_tx = 60, position encoded after origin set — ladder behaving exactly as specified
Crash safety	store + collapses + silence tracks reload after an OS kill (24 h age cutoff)

Table B — comparison with prior art:

Capability	Bridgefy / FireChat	TeamPhone [3]	Epidemic / PRoPHET [1,2]	Zone (ours)
Works with zero infrastructure	Yes	Partial (uses cellular when up)	Yes (theory)	Yes
No pairing / no GATT connection	No (GATT)	Yes	n/a	Yes (connectionless adv.)
Works when the user is unconscious	No	No (needs grouping)	No	Yes (sensor-derived status)
Dual-PHY (Coded for range)	No	No	No	Yes
Battery-honest duty cycle	No	Partial	No	Yes (ladder + honest next_expected_tx)
Infers area destruction from silence	No	No	No	Yes (Dead Man's Packet + CELL_LOSS)
Vertical (floor) triage	No	No	No	Yes (relative barometry)
Spoof / Sybil resistance	Broken [7]	Limited	None	Yes (hashed ID, auth-shape, plausibility, key-verified responder)
Confidence-scored situational map	No	No	No	Yes

Suggested graphs for the slide:

- Line graph — replayed toll curve (deaths vs. hours: 22, 95, 469, 626) with map "cells collapsed" overlaid, showing the map converging in ~90 s of stage time.
- Bar chart — advertised interval and active PHY versus battery % (1 s/Coded, 10 s/alt, 60 s/1M, 300 s/1M).
- Timeline — T3 versus T4: "device powered off" marker, then time-to-classification, one bar ending in UNEXPECTED_SILENCE (escalate) and one in EXPECTED_SILENCE (quiet).

7.3 Discussion of outcomes

- The USP is validated end-to-end.** T3 versus T4 on real hardware proves the core claim — the network distinguishes an announced silence (dying battery) from a catastrophic one (device destroyed), which no prior mesh system does.
- The range / energy trade-off is real and managed.** Literature puts Coded PHY at ~300 m urban and > 1 km ideal [5]; Zone spends that reach only while battery allows, then retreats to 1M — keeping next_expected_tx truthful at every rung.
- "Search this block" becomes "search this 100 m cell."** CELL_LOSS plus confidence scoring convert fragmented reports into a ranked, timestamped picture; the barometric delta adds the vertical axis GPS cannot [6].
- Security posture.** Against the Bridgefy failure modes [7], Zone removes plaintext social graphs (broadcast-only, hashed IDs), rejects malformed frames at the relay, and makes a fabricated-node flood unable to move the map.
- Limitations (stated honestly).** 2.4 GHz penetrates rubble poorly — mitigated by days-long low-duty advertising until a rescuer walks within ~10 m; iOS cannot advertise in the background (Apple restriction); auth is a truncated MAC, not per-device Ed25519 (a deliberate scope cut); full three-phone RESOLVE-propagation and a 30-minute soak test remain to be run.

- **Future work.** LoRa / satellite bridge on the responder phone for the last mile; per-device signatures; full-app localisation; standard GeoJSON export for agency GIS.

References

1. A. Vahdat and D. Becker, "Epidemic Routing for Partially Connected Ad Hoc Networks," Technical Report CS-2000-06, Duke University, 2000.
2. A. Lindgren, A. Doria and O. Schelén, "Probabilistic Routing in Intermittently Connected Networks," Proc. ACM MobiHoc 2003 (poster); formalised as IRTF RFC 6693, 2012.
3. Z. Lu, G. Cao and T. La Porta, "TeamPhone: Networking Smartphones for Disaster Recovery," IEEE Transactions on Mobile Computing, vol. 16, no. 12, pp. 3554–3567, 2017.
4. S. M. Darroudi and C. Gomez, "Bluetooth Low Energy Mesh Networks: A Survey," Sensors (MDPI), vol. 17, no. 7, art. 1467, 2017.
5. A. R. Sheikh et al., "Adaptive Physical Layer Selection for Bluetooth 5: Measurements and Simulations," Wireless Communications and Mobile Computing, vol. 2021, art. 8842919, Wiley/Hindawi.
6. W. Falcon and H. Schulzrinne, "Predicting Floor-Level for 911 Calls with Neural Networks and Smartphone Sensor Data," arXiv:1710.11122, 2017.
7. M. R. Albrecht, J. Blasco, R. B. Jensen and L. Mareková, "Mesh Messaging in Large-Scale Protests: Breaking Bridgefy," in Topics in Cryptology – CT-RSA 2021, LNCS vol. 12704, Springer; IACR ePrint 2021/214.
8. T. Watanabe et al., "Delay-Tolerant Networking for Tsunami Evacuation on the Small Island of Hachijojima: A Study of Epidemic and Prophet Routing," arXiv:2601.00109, 2026.
9. "The earthquake impact on telecommunications infrastructure in Nepal: a preliminary spatial assessment," International Journal of Disaster Risk Reduction (Elsevier), 2023.